Elvis Aguero

MSc. in Applied Mathematics

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Education

2020-2024 **B.Sc., Engineering Physics**, *Universidade Federal da integração Latino-Americana (UNILA)*, Foz do Iguaçu, Brazil. GPA: 3.8/4.0. Summa cum Laude (Expected).

2017–2019 **M. Sc., Applied Maths**, *Instituto de Matemática Pura e Aplicada*, Rio de Janeiro, Brazil.
GPA: 3.6/4.0. Advisor: André Nachbin

Scientific Work

2022 E. A. Aguero, L. Alventosa, D. M. Harris, and C. A. Galeano-Rios. **Impact of a rigid sphere onto an elastic membrane**, *Proceedings of the Royal Society A*, https://doi.org/10.1098/rspa.2022.0340 .

Experience

2017-Present **Olympiad Organizer and Trainer**, *OMAPA*, Asunción, Paraguay (Remote).
- Author of more than 200 math problems for the Paraguayan National Math Olympiad, as well as international math competitions.
- Instructor for the Young Talents Program, where the best students from the National Olympiad receive mathematical and scientific training for international competitions.

2023 **Research Intern**, *Brown University*, Providence, RI.
- Executed fluid dynamics experiments under the mentorship of Prof. Daniel Harris, focusing on deformable droplet impact onto solid surfaces. Developed MATLAB, Julia and Python simulations to validate our experimental findings.

2023 **Scanning Electron Microscope Lab Assistant**, *UNILA*, Foz do Iguaçu, Brazil.
- Provided administrative assistance and prepared samples for material characterization for the Scanning Electron Microscope under the guidance of Prof. Rodrigo Basso

2020-2023 **Academic Consultant**, *Piensa Paraguay*, Ciudad del Este, Paraguay.
- Designed the math curriculum and devised the structure and maintenance of a database with more than 1000 math problems in MongoDB. Piensa is a local startup which offers math, robotics and programming courses for students aged $7-17$.

2019 **Project Intern**, *Parque Tecnológico Itaipu*, Hernandarias, Paraguay.
- Implemented a support vector machine algorithm in MATLAB for classification of faulty high-voltage circuit breakers.

Talks, Teaching and other events

2023-2024 **7th Summer School in Advanced Experimental Physics** organized by the Brazilian Center for Research in Phyics (Acceptance Rate: $\approx 15\%$). Using Ferromagnetic rosonance, magnetron sputtering and x-ray reflectivity, we grew thin bilayer films of ferromagnetic materials to study energy dissipation in the form of coheren angular momenta and spin current.

2023 Contributed talk at the Brazil-China Joint Mathematical Meeting.

2021-2022 Teaching assistant of Numerical Calculus and Physics II at UNILA.

2021 Speaker at the Paraguayan Mathematical Congress.

2021 Speaker at the series of talks "Hugo Checo, creando redes". **Best speaker award**.

2019 Teaching assistant of Functional analysis in the summer school at IMPA.

Programming Languages

Experienced

- MATLAB: Scientific Computing, OOP, Parallel computing, GPU-accelerated algorithms.
- Python: Scientific Computing, OOP, and Basic ML engineering with Pytorch.
- Git, Github: Remote and local version control, contributed to a number of open-source projects.

Familiar

- C++: OOP, Basic Scientific Computing.
- Julia: Scientific Computing, OOP, GPU-accelerated algorithms.
- Java, JS, Google Apps Script, SQL, MongoDB, Google Sheets, Access, LabView, R, bash, HTML, Scilab.

Fellowships and Awards

2017 CNPQ Scholarship for M.Sc. Studies at IMPA.

2017 **Prize "Domingo Martinez de Irala"**: Order of merit Medal, Asunción, Paraguay.

2015, 2016 **International Math Olympiad**: Bronze Medals (2015, 2016).

2015, 2016 **IberoAmerican Math Olympiad**: Honorific Mention (2015), Bronze Medal (2016).

2011-2016 **Paraguayan Math Olympiad**: Three Gold Medals (2013, 2015, 2016) and three Silver Medals (2011, 2012, 2014).

2016 **Distinguished Citizen of Ciudad del Este Medal**.

Volunteer Work and Outreach

2017-Present **Math Olympiads Coordinator, Jury member, Team Leader**, *OMAPA*.

- Part of the Paraguayan delegation as organizer in numerous International Olympiads, such as the Cyberspace Mathematical Competition (2020), IberoAmerican Olympiad (2021), and the International Mathematical Olympiad (2022), which was held in Oslo, Norway.
- Part of the jury committee at the XXXII Cono Sur Olympiad (2021), organized by Paraguay. Graded problems of all countries, and created problems for the contest.

2022-2023 **Engineering Physics Student Council**, *UNILA*.

2019-2022 **Board member, Teaching Staff**, *Mathura*.

- Board member of Mathura, an NGO which gives free Maths and Spanish classes to underserved high school students in their pursuit to undergraduate studies. Also, part of the math teaching staff of Mathura.

2019-2021 **Volunteer Data Analyst**, *ReAcción Paraguay*.

- ReAcción is a non-profit NGO that uses civic education and grassroots mobilization to demand accountability from government in the education sector.
- Used Python and Apps Script to analyze thousand of public contracts, comparing public expenditure of central and local institutions before and after the covid-19 outbreak. A couple of articles (in Spanish) were made based on our findings: https://reaccion.org.py/publicaciones/#investigaciones
- Gave seminars to the public to raise awareness on the importance of open data, and worked on a project to automate the process of contract data analysis in our country.

Languages

Spanish	CEFR C2	Portuguese	CEFR C2
English	CEFR C1	French	CEFR A1

Other Activities

- Problem solving: My ProjectEuler account is percentile 0.4% among all problem solvers.
- When I'm not solving problems, I am usually playing some chess on Lichess!.
- When I'm not playing chess, I am usually running or playing football.